Assignment 10 - Practice Lab - Control Access to Files - Permissions in RHEL 9

Gain hands-on experience with file permissions in RHEL 9.

To complete this practice lab assignment, you need to have access to RHEL 9.

Questions for this assignment

1. **Write the steps for the below task:**
   Task 1: Create and Inspect a File
2. **Write the steps for the below task:**
   Task 2: Modify File Permissions
3. **Write the steps for the below task:**
   Task 3: Set Specific Permissions
4. **Write the steps for the below task:**
   Task 4: Change File Owner
5. **Write the steps for the below task:**
   Task 5: Remove Write Permission
6. **Task 6: Create a New User and Test Permissions**

Solution by the Instructor:

1. Write the steps for the below task:

   Task 1: Create and Inspect a File

   Certainly! Below is a more comprehensive practice lab assignment on the topic of "Control Access to Files - Permissions in Red Hat Enterprise Linux 9 (RHEL 9)" along with answers.

   Practice Lab Assignment: Control Access to Files - Permissions in RHEL 9

   Objective:

   Gain hands-on experience with file permissions in RHEL 9.

   Tasks:

   Task 1: Create and Inspect a File

   1. Open a terminal in RHEL 9.

2. Create a new directory named permissions_lab:

   mkdir permissions_lab

3. Change into the permissions_lab directory:

   cd permissions_lab

4. Create a new file named sample.txt using the following command:

   touch sample.txt

5. Use the ls -l command to inspect the file's permissions, owner, and group.

Expected Output:

-rw-r--r-- 1 your_username your_groupname 0 Jan 01 00:00 sample.txt

Ensure to replace your_username and your_groupname with your actual username and group.

2. Write the steps for the below task:

   Task 2: Modify File Permissions

   1. Grant read and write permissions to the owner of the file sample.txt:

      chmod u+rw sample.txt

   2. Verify the changes using ls -l:

      ls -l sample.txt

   Expected Output:

   -rw-r--r-- 1 your_username your_groupname 0 Jan 01 00:00 sample.txt

   Explain why the permissions did not change as expected.

3. Write the steps for the below task:

   Task 3: Set Specific Permissions

   1. Set the permissions of sample.txt to rw-rw-r--:

      chmod 664 sample.txt

   2. Verify the changes using ls -l:

ls -l sample.txt

Expected Output:

-rw-rw-r-- 1 your_username your_groupname 0 Jan 01 00:00 sample.txt

Confirm that the permissions are now set as intended.

4. Write the steps for the below task:

   Task 4: Change File Owner

   1. Change the owner of sample.txt to the root user:

      sudo chown root sample.txt

   2. Verify the changes using ls -l:

      ls -l sample.txt

   Expected Output:

   -rw-rw-r-- 1 root your_groupname 0 Jan 01 00:00 sample.txt

   Confirm that the owner has changed to the root user.

5. Write the steps for the below task:

   Task 5: Remove Write Permission

   1. Remove write permission from the group for sample.txt:

      chmod g-w sample.txt

   2. Verify the changes using ls -l:

      ls -l sample.txt

   Expected Output:

   -rw-r--r-- 1 root your_groupname 0 Jan 01 00:00 sample.txt

   Confirm that the write permission for the group has been removed.

6. Task 6: Create a New User and Test Permissions

   1. Create a new user named labuser:

      sudo useradd labuser

2. Switch to the labuser account:

   su - labuser

3. Attempt to modify sample.txt:

   echo "Hello, world!" > sample.txt

Observe the error message. Explain why the operation failed.